

Page Layout

1. Global Header & Control Bar

- **Title & Subheading:** Shows "ACME Global Compensation Dashboard" and active database sync status.
- **Global Currency Normalization Selector:** Dropdown toggle to display analytics and payroll metrics normalized in USD, EUR, or GBP (consuming backend exchange rates).
- **Batch Operations Actions:**
 - **"Import CSV" Button:** Triggers the bulk upload dialog for spreadsheet ingestion.
 - **"Add Employee" Button:** Opens the single employee creation modal.

2. Top Section: Compensation Analytics ("How ACME Pays")

KPI Metrics Strip

- Consumes GET /api/analytics/summary.
- **Total Annual Payroll:** Normalized total with month-over-month indicator.
- **Average Base Salary & Median Salary:** Central tendency indicators in selected base currency.
- **Active Headcount:** Total employee count (across Active, On Leave, Terminated statuses).
- **Currency Breakdown:** Mini badge strip displaying distribution across operating currencies (USD, EUR, GBP, etc.).

Visualizations Panel

- **Departmental Distribution (Horizontal Bar Chart - Recharts):** Consumes GET /api/analytics/by-department to show total payroll expenditure and average salary per department.
- **Geographic Compensation Breakdown (Grouped Donut/Bar Chart):** Consumes GET /api/analytics/by-country displaying headcount and spend per operating country.
- **State Handling:** Skeleton card place-holders during query fetching and error retry boundary if analytics endpoints fail.

3. Bottom Section: Server-Driven Employee Data Grid

Toolbar Controls

- **Debounced Search Input:** Text search by employee name or email with 300ms debounce piped to URL query params (?search=...).
- **Faceted Multi-Select Dropdowns:** Filter by Department, Country, Employment Type (Full-time, Part-time, Contractor), and Status (Active, On Leave, Terminated).
- **Export Trigger:** Direct streaming download button for filtered results via GET /api/employees/export.

Data Table (TanStack Table + shadcn/ui)

- **Columns:**
 1. *Employee Name & Email*
 2. *Job Title & Department*
 3. *Country & Office Location*
 4. *Employment Type & Status Badge*
 5. *Local Base Salary & Bonus* (Native Currency)
 6. *Normalized Salary* (USD equivalent)
 7. *Actions Menu* (Edit, View History, Soft Delete)
- **Server-Side Pagination:** Integrated page selector, rows-per-page dropdown (10, 25, 50, 100), and total record count synced to URL search parameters (?page=1&limit=25).
- **Visual Styling:** Sticky action column on the right with a horizontally scrolling container to

support smaller screens without table truncation.

4. Interactive Modals & Workflow Overlays

- **Add / Edit Employee Modal (employee-modal.tsx):**
 - Form fields matching Pydantic backend models: name, email, job_title, department, country, employment_type, status, currency, base_salary, and bonus.
 - Validation schema using react-hook-form + zod strictly validating positive numbers, standard ISO currency codes, and email strings.
 - Displays a side-by-side **Salary Diff Preview** during edits (e.g., "\$120,000 → \$135,000 (+12.5%)") before saving.
- **Batch CSV Ingestion Modal (csv-import-modal.tsx):**
 - Drag-and-drop file uploader accepting .csv and .xlsx files.
 - In-browser schema validation preview parsing the first 5 rows and highlighting malformed emails or negative salary values before dispatching POST /api/employees/batch-import.
- **Salary History Sheet (salary-history-sheet.tsx):**
 - Sliding slide-over drawer showing historical compensation adjustments, effective dates, and audit log notes for the selected employee.
- **Soft-Delete Confirmation Alert (delete-dialog.tsx):**
 - Destructive confirmation dialog triggering DELETE /api/employees/:id with optimistic UI update and Sonner toast rollback on failure.

UI Prompt

AI Generation Prompt: Global Header & Control Bar

Design a desktop UI component for the "Global Header & Control Bar" of an enterprise HR application called ACME Global Compensation Dashboard.

1. Product & Persona:

- *It is an internal salary management tool used exclusively by an HR Manager managing compensation for 10,000 global employees across multiple countries.*

2. What Must Appear (Key Features):

- **Left Side:** Clean product branding reading **"ACME Compensation"** with a sub-label or badge indicating **"Global Payroll Sync • Active"**.
- **Right Side Controls:**
 - A **Global Currency Selector** (segmented control or select dropdown toggling between **USD (\$)**, **EUR (€)**, and **GBP (£)**) with a subtle label: "Base Currency".
 - A secondary action button: **"Import CSV"** (with an upload/spreadsheet icon).
 - A primary CTA button: **"+ Add Employee"**.

3. What NOT to Include (Boundaries):

- *Do not add public SaaS marketing navigation links (e.g., Pricing, Solutions, Docs, Contact).*

- Do not add random notification bells, search bars (the search belongs in the data grid below), or floating gradient elements.
- Avoid glowing AI aesthetics, neon buttons, or oversized empty padding.

4. Visual Direction:

- Make it look like expensive, restrained enterprise software (similar to Linear or shadcn/ui).
- Compact spacing, subtle 1px border on the bottom, clean typography, neutral dark or slate-gray accents, and clear visual hierarchy.

Interpret this direction and generate the interface design for this top control bar.

Follow-Up Refinement

"The layout is solid. Keep the left branding and currency selector, but make the button styling more restrained and decrease vertical padding to give it higher data density."

AI Generation Prompt: "Top Section: Compensation Analytics ("How ACME Pays")"

Design a desktop UI analytics section titled "How ACME Pays" for an internal enterprise compensation management tool.

1. Product & Purpose:

- This section provides an HR Manager with an instant, high-level view of how a 10,000-employee global organization distributes salaries and headcount across departments and operating countries.

2. What Must Appear (Key Components):

- **Row 1 — KPI Metrics Strip (4 Clean Stat Cards):**
 - **Total Annual Payroll:** A prominent currency value (e.g., \$142.5M USD) with a subtle subtitle for normalized total spend.
 - **Average Base Salary:** Mean compensation value (e.g., \$98,400 USD).
 - **Median Salary:** Midpoint benchmark (e.g., \$92,000 USD).
 - **Active Headcount & Currencies:** Total employee count (e.g., 10,000 Employees) with small pills/badges for operating currencies (USD, EUR, GBP, CAD).
- **Row 2 — Analytical Visualizations (2 Side-by-Side Chart Cards):**
 - **Left Card (Departmental Breakdown):** A horizontal bar chart showing total payroll budget and average compensation across key departments (Engineering, Sales, Product, Marketing, Operations, HR).
 - **Right Card (Geographic Spend Distribution):** A grouped bar or clean donut chart displaying total compensation spend and headcount split

across operating countries (US, UK, Germany, Canada, Nigeria).

3. What NOT to Include (Boundaries):

- Do not include individual employee lists, tables, or row-level records (those belong in the table section below).
- Do not include vague vanity metrics (no Website Traffic, Conversion Rates, Click-Throughs, or Satisfaction Scores).
- Avoid bright neon color schemes, glowing AI mesh gradients, or 3D chart graphics.

4. Visual Direction:

- High data density, quiet, restrained, and enterprise-grade (inspired by shadcn/ui and modern financial dashboards).
- Clean card containers with subtle 1px borders, neutral background fills, clear typography hierarchy, and precise muted chart accents.

Interpret this direction and generate the visual UI design for this analytics dashboard section.

Follow-Up Refinement

"Keep the 4 KPI cards and the horizontal bar chart. Adjust the right chart into a simpler, grouped distribution card and reduce card padding to maintain high information density."

Prompt for Generating "Bottom Section: Server-Driven Employee Data Grid"

Design a desktop UI data table section for managing a 10,000-employee compensation directory in an enterprise HR application.

1. Product & Purpose:

- This section provides an HR Manager with an interactive, server-paginated data grid to search, filter, view dual-currency compensation, and manage individual employee records.

2. What Must Appear (Key Components):

- **Toolbar Controls (Top Strip):**
 - **Search Bar:** Debounced search input with a placeholder: "Search by employee name, title, or email...".
 - **Faceted Multi-Select Filter Dropdowns:** Compact dropdown filters for

Department, Country, Employment Type (Full-time, Part-time, Contractor), and Status (Active, On Leave, Terminated).

- **Actions:** A secondary "Export CSV" button (with a download icon).
- **Data Table Columns (High-Density Grid):**
 - **Employee:** Full Name & Email (stacked sub-text).
 - **Title & Dept:** Job Title & Department pill.
 - **Location:** Country flag/code & City.
 - **Status:** Compact status badges (e.g., green for Active, amber for On Leave, neutral for Terminated).
 - **Local Compensation:** Native Base Salary & Bonus (e.g., £75,000 + £5,000 or €82,000).
 - **Normalized Salary (USD):** Standardized USD total figure (e.g., \$98,250 USD).
 - **Actions (Sticky Column):** Subtle three-dots menu icon (triggering Edit, View History, Soft Delete).
- **Pagination Footer (Bottom Strip):**
 - Left side: "Showing 1–25 of 10,000 records".
 - Right side: Rows-per-page selector (25 / page) and numbered page navigation buttons (< 1 2 3 ... 400 >).

3. What NOT to Include (Boundaries):

- Do not include decorative analytics cards, KPI summary metrics, or large marketing banners (these are handled in sections above).
- Do not make every table row a massive floating card; it must be a structured, dense data table.
- Avoid glowing hover effects, oversized avatars, or cartoonish icons.

4. Visual Direction:

- Clean, dense, and precise enterprise table design (styled like shadcn/ui and TanStack Table).
- Subtle horizontal dividing lines (1px), monospace formatting for numeric salary columns, muted badge colors, and crisp typographic hierarchy.

Interpret this direction and generate the visual UI design for this employee data grid section.

Follow-Up Refinement

"The layout structure is right. Increase data density by reducing vertical cell padding, ensure both local currency and normalized USD are clearly aligned in the salary column, and keep the action menu visible on the right edge."

Prompt for Generating "Interactive Modals & Workflow Overlays"

Design a desktop UI modal and slide-over overlay system for an enterprise salary management application.

1. Product & Purpose:

- These modal overlays allow an HR Manager to perform employee CRUD operations, batch import data from Excel, view compensation change history, and handle record deletions.

2. What Must Appear (Key Overlays to Showcase):

- **Main Focal Component — Add / Edit Employee Modal (`employee-modal.tsx`):**
 - A clean, centered modal dialog with two-column input fields:
 - **Personal & Role:** `Full Name`, `Email`, `Job Title`, `Department`, `Country`, `Employment Type` (Full-time, Part-time, Contractor), and `Status` (Active, On Leave, Terminated).
 - **Compensation Details:** `Currency` (USD, EUR, GBP), `Base Salary`, and `Bonus`.
 - **Salary Diff Comparison Card:** A subtle highlight box inside the edit state showing previous vs. updated salary (e.g., "Current: \$120,000 → Proposed: \$135,000 (+12.5%)").
 - Primary CTA: **"Save Changes"** and a neutral **"Cancel"** button.
- **Secondary Overlays (Shown as supporting context or open drawer):**
 - **Batch CSV Ingestion Dialog (`csv-import-modal.tsx`):** Drag-and-drop file upload zone with a tabular 3-row validation preview (flagging valid rows in green and malformed values in red).
 - **Salary History Drawer (`salary-history-sheet.tsx`):** A right-side slide-over sheet showing an audit timeline of past compensation adjustments and effective dates.
 - **Destructive Delete Dialog (`delete-dialog.tsx`):** A compact confirmation alert with a red **"Delete Employee"** action.

3. What NOT to Include (Boundaries):

- Do not include arbitrary multi-step onboarding wizards or marketing upsell modals.
- Do not make modal backgrounds opaque without a dark semi-transparent backdrop blur.
- Avoid bright, flashy color schemes or cartoonish error illustrations.

4. Visual Direction:

- *Restrained, modern enterprise design (styled with shadcn/ui Dialog and Sheet conventions).*
- *High form density, crisp input borders, neutral gray backgrounds, subtle elevation shadows, clear validation states, and standard system typography.*

Interpret this direction and generate the visual UI design for these interactive workflow modals and overlays.

Follow-Up Refinement

"Focus exclusively on the Add/Edit Employee Modal and the right-side Salary History Drawer. Keep the two-column form layout and ensure the salary diff comparison box is visually distinct."